Machine Learning foundation labs

Four two-hour guided labs, one on a fourth teaching day in each of weeks 1 to 4. They are guided practice, not extra homework. Each lab follows its week's third core session and precedes the next week. Keep two 45 to 60 minute practice blocks on non-class days.

Week 1: ML03, then F01

Numbers, names and predicting Python output. Output: F01.ipynb with a worked result, an individual variation and a plain-language explanation.

Week 2: ML06, then F02

Lists, vectors and NumPy arrays. Output: F02.ipynb with a worked result, an individual variation and a plain-language explanation.

Week 3: ML09, then F03

Variation, proportions and probability. Output: F03.ipynb with a worked result, an individual variation and a plain-language explanation.

Week 4: ML12, then F04

Matrices, weighted sums and slope. Output: F04.ipynb with a worked result, an individual variation and a plain-language explanation.

Readiness before ML13

Each learner explains a variable, indexes a list and an array, reads a row/column table, calculates a mean and a proportion, and works one weighted sum with units. Check individually. Use a repair example if a skill is missing. Attendance is not understanding.